

1. The derivative as a function

Recall that a function defined on an open interval containing a is **differentiable at a** if the finite limit

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists. As a varies, these values define a new function, the **derivative** f' . Its domain consists of the interior points of the original domain at which this finite, two-sided limit exists. An infinite limit does not give a derivative under this definition.

For $f(x) = x^2$, the last lecture showed $f'(a) = 2a$ for every $a \in \mathbb{R}$. Thus $f'(x) = 2x$, on $\mathbb{R}$. We also write $\frac{d}{dx}f(x)$ for $f'(x)$.

Example

If $f(x) = c$, then $(f(a+h) - f(a))/h = 0$ for $h \neq 0$, so $f'(a) = 0$. If $f(x) = mx + b$, then

$$\frac{f(a+h) - f(a)}{h} = \frac{m(a+h) + b - (ma + b)}{h} = m, \quad f'(a) = m.$$

Both derivatives are defined on $\mathbb{R}$. The slope of the tangent to a line is the slope of the line itself.

2. Differentiability and continuity

Theorem. If f is differentiable at a , then f is continuous at a .

Proof. For $h \neq 0$,

$$f(a+h) - f(a) = h \frac{f(a+h) - f(a)}{h}.$$

As $h \rightarrow 0$, the two factors tend to 0 and the finite number $f'(a)$. By the product law,

$$\lim_{h \rightarrow 0} [f(a+h) - f(a)] = 0.$$

Hence $\lim_{x \rightarrow a} f(x) = f(a)$, as required. $\square$

Example

The converse is false. The function $f(x) = |x|$ is continuous at zero, since $|f(x) - f(0)| = |x|$, so $\delta = \varepsilon$ works in the continuity definition. But

$$\frac{f(h) - f(0)}{h} = \frac{|h|}{h} = \begin{cases} 1, & h > 0, \\ -1, & h < 0. \end{cases}$$

The two one-sided limits disagree. Thus $f'(0)$ does not exist. For $x > 0$, the function agrees locally with x ; for $x < 0$, it agrees locally with $-x$. Therefore

$$f'(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0, \end{cases} \quad \text{with domain } \mathbb{R} \setminus \{0\}.$$

The original function has domain $\mathbb{R}$; its derivative has a smaller domain. A discontinuity rules out differentiability immediately; continuity alone does not settle the question.

3. Powers and linearity

Positive integer powers. If n is a positive integer, then

$$\boxed{\frac{d}{dx}x^n = nx^{n-1}.$$

For $n = 1$, this is the derivative of the line $y = x$. For $n \geq 2$, the following proof works at every $a \in \mathbb{R}$, including $a = 0$.

Proof. Factor the difference of powers. For $x \neq a$,

$$\frac{x^n - a^n}{x - a} = x^{n-1} + x^{n-2}a + \cdots + xa^{n-2} + a^{n-1}.$$

There are n terms, each tending to a^{n-1} as $x \rightarrow a$. The sum law gives

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}.$$

This is the derivative at a . $\square$

Linearity. Suppose f and g are differentiable at a , and c, d are real constants. Then $cf + dg$ is differentiable at a , and

$$\boxed{(cf + dg)'(a) = cf'(a) + dg'(a).$$

Proof. For nonzero h sufficiently close to zero,

$$\frac{(cf + dg)(a + h) - (cf + dg)(a)}{h} = c \frac{f(a + h) - f(a)}{h} + d \frac{g(a + h) - g(a)}{h}.$$

Take limits and use the scalar and sum laws. $\square$

In particular, $(f + g)' = f' + g'$, $(f - g)' = f' - g'$, and $(cf)' = cf'$, at points where the indicated derivatives exist. Repeated application gives derivatives of finite sums. For a polynomial,

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

we obtain

$$p'(x) = na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \cdots + a_1.$$

Thus every polynomial is differentiable everywhere on $\mathbb{R}$.

Example

For $p(x) = x^3 - 3x^2 + 2$,

$$p'(x) = 3x^2 - 6x = 3x(x - 2).$$

The tangent is horizontal when $p'(x) = 0$, namely at $x = 0$ and $x = 2$. Their points are $(0, 2)$ and $(2, -2)$, and the tangent lines are $y = 2$ and $y = -2$.

4. Products and quotients

Assume f and g are defined on an open interval containing a and differentiable at a .

Product rule. The product is differentiable at a , with

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

Proof. Add and subtract $f(a+h)g(a)$ in the numerator:

$$\frac{f(a+h)g(a+h) - f(a)g(a)}{h} = f(a+h)\frac{g(a+h) - g(a)}{h} + g(a)\frac{f(a+h) - f(a)}{h}.$$

Differentiability makes f continuous at a , so $f(a+h) \rightarrow f(a)$. Taking limits proves the formula. $\square$

Reciprocal rule. If also $g(a) \neq 0$, then

$$\left(\frac{1}{g}\right)'(a) = -\frac{g'(a)}{g(a)^2}.$$

Proof. Continuity of g at a gives $r > 0$ such that

$$|x - a| < r \implies |g(x) - g(a)| < \frac{|g(a)|}{2}, \quad |g(x)| > \frac{|g(a)|}{2} > 0.$$

Thus the reciprocal is defined near a . For $0 < |h| < r$,

$$\frac{\frac{1}{g(a+h)} - \frac{1}{g(a)}}{h} = -\frac{g(a+h) - g(a)}{h} \frac{1}{g(a+h)g(a)} \rightarrow -\frac{g'(a)}{g(a)^2}.$$

Here we used continuity and the quotient limit law. $\square$

Quotient rule. Under the same hypotheses, with $g(a) \neq 0$, apply the product rule to $f \cdot (1/g)$:

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)}{g(a)} - \frac{f(a)g'(a)}{g(a)^2} = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

In particular, a rational function is differentiable wherever its denominator is nonzero.

Negative integer powers. For a positive integer m and $x \neq 0$,

$$\frac{d}{dx}x^{-m} = \frac{d}{dx}\frac{1}{x^m} = -\frac{mx^{m-1}}{x^{2m}} = -mx^{-m-1}.$$

Together with the positive-power and constant rules, this handles integer powers. The constant function 1 has derivative zero.

Example

For $r(x) = (x^2 + 1)/(x - 1)$, on $\mathbb{R} \setminus \{1\}$,

$$r'(x) = \frac{2x(x-1) - (x^2+1)}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}.$$

5. Trigonometric derivatives

Angles are measured in **radians**. From the last lecture,

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1, \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0.$$

The second limit follows by changing the sign of $(1 - \cos h)/h$. These are the two inputs for the following proofs.

Sine. At any $a \in \mathbb{R}$, the addition formula gives

$$\begin{aligned} \frac{\sin(a+h) - \sin a}{h} &= \frac{\sin a \cos h + \cos a \sin h - \sin a}{h} \\ &= \sin a \frac{\cos h - 1}{h} + \cos a \frac{\sin h}{h}. \end{aligned}$$

Taking limits yields $\sin a \cdot 0 + \cos a \cdot 1 = \cos a$. Hence

$$\boxed{\frac{d}{dx} \sin x = \cos x \quad (x \in \mathbb{R}).}$$

Cosine. Similarly,

$$\begin{aligned} \frac{\cos(a+h) - \cos a}{h} &= \frac{\cos a \cos h - \sin a \sin h - \cos a}{h} \\ &= \cos a \frac{\cos h - 1}{h} - \sin a \frac{\sin h}{h} \longrightarrow -\sin a. \end{aligned}$$

Therefore

$$\boxed{\frac{d}{dx} \cos x = -\sin x \quad (x \in \mathbb{R}).}$$

Example

For $f(x) = x \sin x$, the product rule gives

$$f'(x) = \sin x + x \cos x.$$

At $x = \pi$, the point is $(\pi, 0)$ and the slope is $f'(\pi) = -\pi$. Its tangent is

$$y = -\pi(x - \pi).$$

Other trigonometric functions

For $\cos x \neq 0$, the quotient rule and $\sin^2 x + \cos^2 x = 1$ give

$$\frac{d}{dx} \tan x = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

The reciprocal rule gives

$$\frac{d}{dx} \sec x = -\frac{-\sin x}{\cos^2 x} = \sec x \tan x.$$

For $\sin x \neq 0$, the same rules give

$$\frac{d}{dx} \cot x = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\csc^2 x, \quad \frac{d}{dx} \csc x = -\frac{\cos x}{\sin^2 x} = -\csc x \cot x.$$

The formulas, together with their domains, are:

Function	Derivative	Domain of both functions
$\sin x$	$\cos x$	$\mathbb{R}$
$\cos x$	$-\sin x$	$\mathbb{R}$
$\tan x$	$\sec^2 x$	$\cos x \neq 0$
$\sec x$	$\sec x \tan x$	$\cos x \neq 0$
$\cot x$	$-\csc^2 x$	$\sin x \neq 0$
$\csc x$	$-\csc x \cot x$	$\sin x \neq 0$

Here $\cos x = 0$ at $x = \pi/2 + k\pi$, and $\sin x = 0$ at $x = k\pi$, for integers k .

Example

Let $F(x) = (1 + \sin x)/\cos x$, with $\cos x \neq 0$. Then

$$\begin{aligned} F'(x) &= \frac{\cos^2 x + (1 + \sin x) \sin x}{\cos^2 x} \\ &= \frac{1 + \sin x}{\cos^2 x}. \end{aligned}$$

One can also use $F(x) = \sec x + \tan x$: its derivative is $\sec x \tan x + \sec^2 x$, the same expression. In both calculations the domain remains $\cos x \neq 0$.

6. Exponential derivatives

Recall that $\ln$ is the inverse of $x \mapsto e^x$, and $a^x = e^{x \ln a}$ for $a > 0$. In Lecture 2 we introduced

$$e = \sup \left\{ \left(1 + \frac{1}{n} \right)^n : n = 1, 2, 3, \dots \right\}.$$

For this number, we use the following result **without proof**:

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$$

Equivalently, the slope of e^x at zero is 1. Proving this limit from the earlier construction of e requires an additional argument; it is not a consequence of the power laws alone.

Derivative of e^x . For every $x \in \mathbb{R}$,

$$\frac{e^{x+h} - e^x}{h} = e^x \frac{e^h - 1}{h} \longrightarrow e^x. \quad \boxed{\frac{d}{dx} e^x = e^x.}$$

Thus e^x is differentiable, and hence continuous, on $\mathbb{R}$.

A general positive base. Fix $a > 0$, $a \neq 1$, and put $k = \ln a \neq 0$. As $h \rightarrow 0$, $t = kh \rightarrow 0$, so

$$\lim_{h \rightarrow 0} \frac{a^h - 1}{h} = \lim_{h \rightarrow 0} k \frac{e^{kh} - 1}{kh} = k = \ln a.$$

This is a linear change in the variable of a limit. For $k < 0$ the two sides of zero are interchanged; the same two-sided limit applies. Using $a^{x+h} = a^x a^h$, we obtain

$$\frac{a^{x+h} - a^x}{h} = a^x \frac{a^h - 1}{h} \longrightarrow a^x \ln a. \quad \boxed{\frac{d}{dx} a^x = a^x \ln a.}$$

For $a = 1$, this also holds: the function is constant and $\ln 1 = 0$. Every positive-base exponential is therefore differentiable on $\mathbb{R}$.

Example

$$\frac{d}{dx} 2^x = 2^x \ln 2, \quad \frac{d}{dx} \left(\frac{1}{2} \right)^x = - \left(\frac{1}{2} \right)^x \ln 2.$$

The signs agree with the increasing and decreasing exponentials from Lecture 1. The base is fixed and the exponent varies; this is different from x^n , where the exponent is fixed.

Example

For $H(x) = x^2 e^x$, the product rule gives

$$H'(x) = 2x e^x + x^2 e^x = e^x (x^2 + 2x).$$